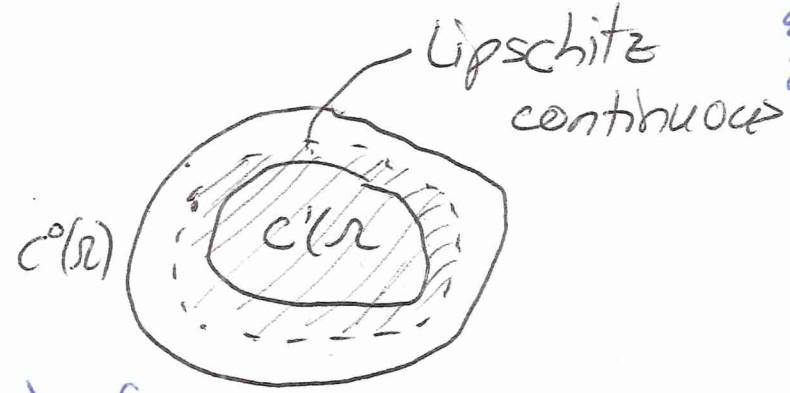
§1.4 ODE Misc

8-28

① Lipschitz ~~conti~~ continuous

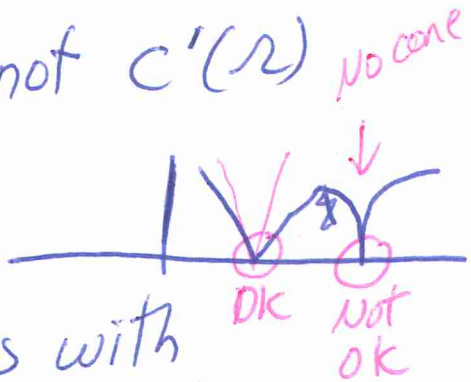


A function $f(x)$ is Lipschitz contin. (or satisfies the Lipschitz condition) if for all x_1, x_2 in a domain (open interval in $\mathbb{R}$) there is a constant $A > 0$ such that

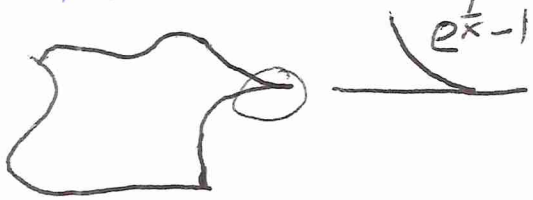
$$|f(x_1) - f(x_2)| \leq A\,|x_1 - x_2|$$

- All functions that are $C^1(\Omega)$ are Lipschitz
  proved last class
- There are functions that are Lipschitz but not $C^1(\Omega)$

ex) $|x|$



~~Take~~ Think: Lipschitz functions are C^1 functions with cone condition, i.e. finite number of corners are ok.



Cone 2D

Lipschitz f'ns are useful in ODE/PDE Thms

ex) Recall existence + uniqueness Thm for 1st-order nonlinear ODE's:

$$x' = f(t,x) \qquad x(t_0) = x_0$$

If there is an open rectangle R such that

if (i) $f(t,x)$ is continuous over rect. R

(ii) $\frac{\partial f}{\partial x}(t,x)$ is Lipschitz → more than $C^0(R)$ with respect to x over R

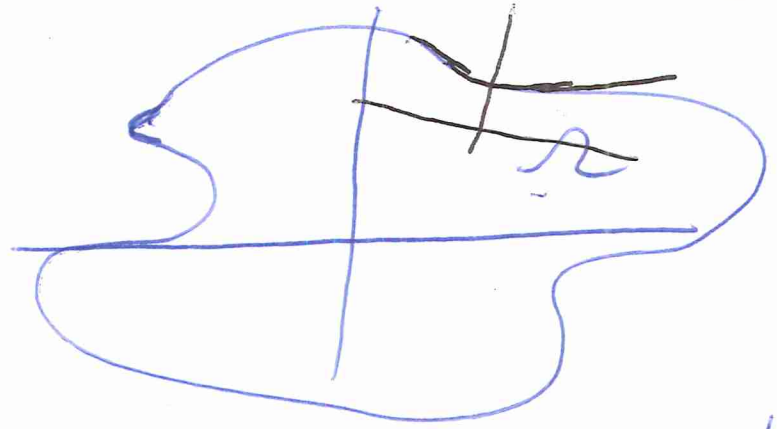
(iii) $(t_0, x_0) \in R$, rect.

Then existence + uniqueness.

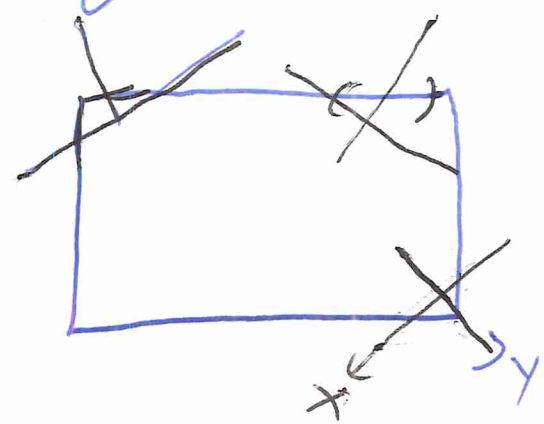
ex) How to define the smoothness of a boundary?

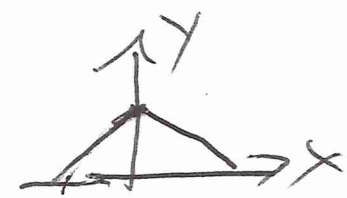
bdry $\partial\Omega$

ex) $\mathbb{R}^2$



we say $\partial\Omega$ is C^k if locally we can put a coord. system and the curve is C^k everywhere





For computations

Not C^1 domain

But is Lipshitz

② Leibniz Rule

combines of Fund Thm of Calc
(integrals & deriv. undo each other)
and chain rule

$$\frac{d}{dx}\int_{g(x)}^{h(x)} f(x,t)\,dt$$

$$= \int_{g(x)}^{h(x)} \frac{\partial}{\partial x} f(x,t)\,dt - f(x,g(x))\,g'(x) + f(x,h(x))\,h'(x)$$

ex) $$\frac{d}{dx}\int_{x^3}^{x^2} \frac{dt}{x+t} = \int_{x^3}^{x^2} \frac{\partial}{\partial x}\left(\frac{1}{x+t}\right)dt + \frac{1}{x+x^2}(2x) - \frac{1}{x+x^3}(3x^2)$$

$$\frac{\partial}{\partial u}\left(\frac{1}{u}\right) = \frac{\partial}{\partial u}u^{-1} = -u^{-2}$$

$$= \int_{x^3}^{x^2} -\frac{1}{(x+t)^2}\,dt + \frac{2x}{x+x^2} - \frac{3x^2}{x+x^3}$$

$$= \frac{1}{x+t}\Big|_{t=x^3}^{x^2} + \frac{2}{1+x} - \frac{3x}{1+x^2} = \frac{1+2x}{x+x^2} - \frac{1+3x^2}{x+x^3}$$

ex) $$\frac{d}{dx}\underbrace{\int_a^x f(t)\,dt}_{F(x)-F(a)} = \int_a^x \cancel{\frac{\partial f(t)}{\partial x}}^{\,0}\,dt + f(x)\cancel{\frac{dx}{dx}}^{\,1} - f(a)\cancel{\frac{da}{dx}}^{\,0}$$

$$= f(x)$$

ex) $$\frac{\partial}{\partial x}\Big|_t \int_{xt}^{x^2t^2} e^{xt-y}\,dy = \int_{xt}^{x^2t^2} \frac{\partial}{\partial x}\left[e^{xt-y}\right]dy$$

treat t as a constant

$$+\left(e^{xt-x^2t^2}\right)\frac{\partial}{\partial x}\left(x^2t^2\right)$$

$$-\left(e^{xt-xt}\right)\frac{\partial}{\partial x}\left(xt\right)$$

$$= \int_{xt}^{x^2t^2} t e^{xt-y}\,dy + e^{xt-x^2t^2}\,2xt^2 - e^0(t)$$

$u = xt-y \quad du = -dy$

$$= -t e^{xt-y}\Big|_{y=xt}^{x^2t^2} + 2xt^2 e^{xt-x^2t^2} - t$$

$$= -t\left[e^{xt-x^2t^2} - e^{xt-xt}\right] + 2xt^2 e^{xt-x^2t^2} - t$$

cancel

$$= -t e^{xt - x^2t^2} + 2xt^2 e^{xt - x^2t^2}$$

$$= -e^{xt - x^2t^2}(t - 2xt^2)$$

$$= e^{xt - x^2t^2}(2xt^2 - t)$$

same ☺

<u>Check</u>

not using Leibniz rule

$$\frac{\partial}{\partial x}\int_{xt}^{x^2t^2} e^{xt-y}\,dy = \frac{\partial}{\partial x}\left[-e^{xt-y}\right]_{y=xt}^{x^2t^2}$$

$$= \frac{\partial}{\partial x}\left[-e^{xt-x^2t^2} + e^{xt-xt}\right]$$

$$= \frac{\partial}{\partial x}\left[-e^{xt-x^2t^2} + 1\right]$$

$$= -e^{xt-x^2t^2}\left[t - 2xt^2\right]$$

For HW over S1.4

solve a system of ODE's

① Linear system with constant coeff

$$\frac{dx_1}{dt} = ax_1 + bx_2 \qquad \frac{dx_2}{dt} = cx_1 + dx_2$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} a & b \\ c & d \end{bmatrix}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Find eigenvalues of matrix, λ_1, λ_2

& eigenvectors

3 cases

① $\lambda_1 \neq \lambda_2$, both real $\qquad \underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = c_1 e^{\lambda_1 t} \underline{v}_1 + c_2 e^{\lambda_2 t} \underline{v}_2$

② $\lambda_1 = \lambda_2$ both real

look it up.

③ λ_1, λ_2 complex conjug. $\qquad \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = e^{\alpha t} \{\cos(\beta t) + i \sin(\beta t)\} \underline{v}$

$\lambda_1 = \alpha + i\beta$

If have complex sol'n, Then The real part + The imag. part are also sol'ns.

② Elimination

x_1 eliminate $\Rightarrow$ solve for x_2